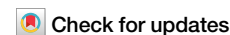


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Teleportation-based speed meter for precision measurement

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We propose a quantum teleportation-based speed meter for interferometric displacement sensing. The primary motivation is to transform a conventional position-sensing interferometer into a quantum non-demolition speed measurement device, without modifying its fundamental optical configuration. Two equivalent implementations are presented: an online approach that uses real-time displacement operation and an offline approach that relies on post-processing. Both implementations reduce quantum radiation pressure noise and surpass the standard quantum limit of measuring displacement, and they can be applied to a wide range of interferometer configurations. We discuss potential applications to gravitational-wave detectors, where our scheme enhances low-frequency sensitivity without requiring modifications to the core optics of a conventional Michelson interferometer (e.g., substrate or coating properties). This approach offers a new path to back-action evasion enabled by quantum entanglement.

Precision measurements in cavity optomechanics are fundamentally limited by quantum uncertainties. In free-mass displacement measurements, Heisenberg's uncertainty principle imposes the standard quantum limit (SQL)¹, arising from a trade-off between measurement precision (associated with position uncertainty) and back-action (associated with momentum uncertainty). Improving measurement precision inevitably leads to an increase in back-action noise.

Various back-action evasion techniques have been developed to mitigate this noise, such as squeezing injection^{2–6}, variational readout^{6,7}, effective negative-mass systems^{8–14}, and optical-spring techniques¹⁵. Among these, squeezed-state injection has become well-established in gravitational-wave detectors (GWDs). By squeezing a specific quadrature of the vacuum field, one can reduce either position or momentum uncertainty at the expense of the other. The Laser Interferometer Gravitational-wave Observatory (LIGO) has demonstrated an improvement of approximately 3 dB at around 50 Hz, exceeding the SQL^{16,17}.

An alternative way to beat the SQL-without relying on squeezing-is to perform quantum non-demolition (QND) measurements of observables that commute with themselves at different times^{18,19}. For a free mass, the momentum operator is one such operator and thus can be monitored in a QND manner that inherently beats the SQL without squeezing. Since momentum is proportional to velocity, such a device that monitors the momentum is often called a speed meter^{20,21}.

Consider a system in which the optical field is coupled to a mechanical mirror, and the motion of the mirror is weakly and continuously monitored. The mirror's natural frequency is assumed to be much smaller than the cavity bandwidth (the so-called bad-cavity regime), and we also assume that the mirrors are allowed to be treated effectively as a free mass within the observation band.

Figure 1a illustrates a setup in which the mirror's position $\hat{x}(t)$ is linearly encoded in the phase of the outgoing light, i.e., $\hat{\phi}(t) \propto \hat{x}(t)$. A Michelson interferometer-used widely for displacement measurement - is an example of such a position meter. In this type of measurement, the sensitivity is ultimately limited by quantum noise, comprising both quantum radiation pressure noise (QRPN) and shot noise.

In such position measurements, we cannot avoid SQL. The single-sided standard quantum limit (SQL) is given by¹:

$$x_{\text{SQL}}(\Omega) = \sqrt{\frac{2\hbar}{m\Omega^2}} \quad (1)$$

where Ω is the angular frequency, $\hbar$ is the reduced Planck constant, and m is the mass of the test object. The SQL defines the fundamental sensitivity limit for classical position measurements, corresponding to the minimum achievable quantum noise as the input laser power is optimized. The SQL arises from the non-commutativity of the position operators at different times; that is, the position operator of the test mass at time t does not

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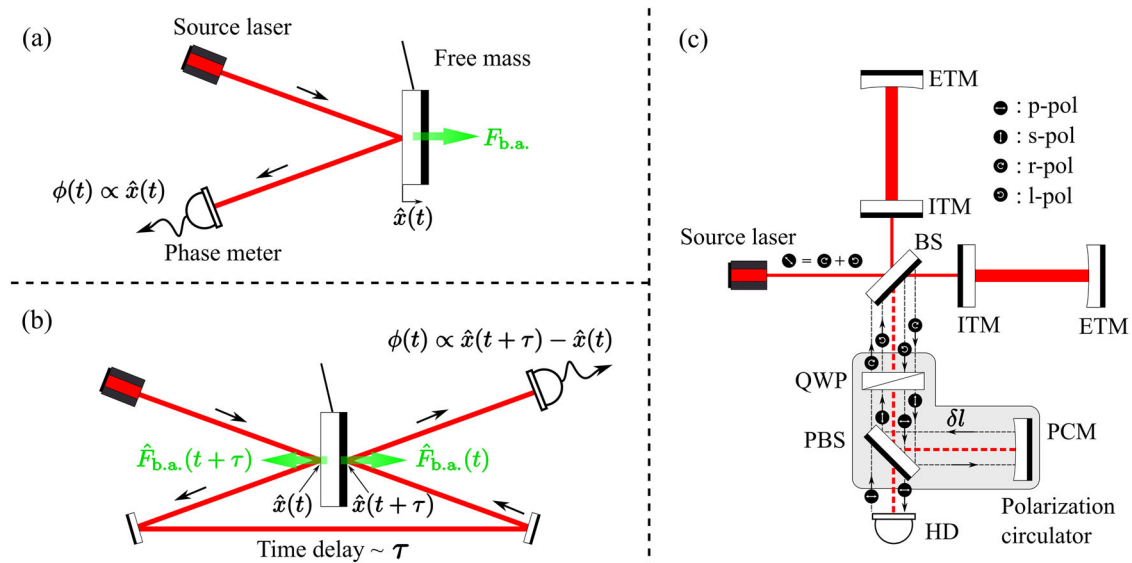


Fig. 1 | Toy models of position and speed meters. **a** Simplified toy model of a position measurement of a free mass. The information about the test mass’s position is imprinted in the phase of the reflected light as $\phi(t) \propto \hat{x}(t)$. **b** Toy model of a speed meter, as presented in Refs. 21,23. The light couples with the mirror twice retaining its coherence. The measured quantity is proportional to the difference between the position information at the time t and $t + \tau$, i.e., $\hat{x}(t + \tau) - \hat{x}(t)$. The backaction forces, $\hat{F}_{b.a.}(t)$ and $\hat{F}_{b.a.}(t + \tau)$, are canceled out with each other by pushing the mirror in the opposite directions. **c** An example of a speed meter: the polarization circulation speed meter (PCSM). The main laser is linearly polarized at 45° , composed of orthogonal right (r) and left (l) circular polarizations. Due to the polarization beam splitter (PBS), only p-polarized vacuum enters the interferometer from the detection port. This p-polarized vacuum is converted into r-polarization by a

quarter-wave plate (QWP), and then couples to the classical amplitude of the main laser. This coupling induces random radiation pressure and causes back-action. The outgoing r-polarized light is converted into s-polarization by the QWP and is recycled back into the interferometer via the PBS and the polarization circulation mirror (PCM). The s-polarized light is further converted into l-polarization by the QWP and couples to the l-polarized component of the main laser. The final outgoing field is converted back into p-polarization, transmitted through the PBS, and detected by the homodyne detector (HD). Abbreviations: ITM (ETM), input (end) test mass; BS, beam splitter. The measurement reveals no information about Victor’s state. Due to the perfect quantum correlations in an ideal EPR state, only Bob can reconstruct Victor’s state using the classical information from Alice.

commute with the one at a later time t' :

$$[\hat{x}(t), \hat{x}(t')] \neq 0 \quad (2)$$

due to the back action force $\hat{F}_{b.a.}(t)$. (equivalently momentum $\hat{p}_{b.a.}(t)$. impinging on the mirror. This force is the source of the QRPN. Consequently, there is an inherent trade-off between QRPN and shot noise, which prevents both noise sources from being simultaneously reduced.

In contrast, observables that commute at different times can be monitored with arbitrary precision. These are referred to as QND observables. The momentum of a free mirror is one example of a QND observable. For a freely evolving mass, its momentum is proportional to its speed, i.e., $\hat{p} = m\hat{v}$, which is why a momentum meter is also called a “speed meter.” However, when the mirror is coupled to a probe such as light, a more careful analysis reveals that the canonical momentum is no longer simply proportional to the speed. Instead, it takes the form

$$\hat{p} = m\hat{v} - \alpha\hat{a}_1, \quad (3)$$

where $\hat{a}_1$ is the amplitude quadrature of the probe light and α represents the optomechanical coupling strength. Nevertheless, it is still possible to reduce the back-action noise below the SQL by exploiting the correlation between the phase and amplitude quadratures of the outgoing field (see Section 2.11 of Ref. 22 and Section 4.5.2 of Ref. 23).

An illustration of the light-mirror interaction is provided in Fig. 1b (see also Refs. 21,23). Light from the source laser impinges on the front side of the movable mirror, is reflected, and then recycled to the rear side of the mirror with a time delay τ . At the second interaction (at time $t + \tau$), the intensity fluctuations of the light push the mirror with $\hat{F}_{b.a.}(t + \tau)$ in the opposite direction compared to the first interaction, $\hat{F}_{b.a.}(t)$ at time t , thereby canceling the initial radiation pressure force. Consequently, the observable

measured by the phase meter becomes

$$\phi \propto \hat{x}(t + \tau) - \hat{x}(t) \sim \tau\bar{v} \quad (4)$$

where $\bar{v}$ denotes the average velocity of the mirror. Note that when the light interacts with the mirror inside optical cavities, the delay τ corresponds to the cavity storage time.

Many configurations of speed meters have been proposed in the context of gravitational-wave detection^{24–27}. Figure 1c: illustrates the design of the polarization circulation speed meter (PCSM)²⁶. Instead of using the front and rear sides of the optical modes, this proposal employs two orthogonal polarization modes coexisting in the same arm cavities. By tuning the phase of the orthogonal polarization and adjusting several polarization optics at the detection port, one can effectively cancel out the measurement back-action (for details, see the caption of Fig. 1c and Ref. 26).

The PCSM is regarded as the most feasible candidate for future GWDs, since it does not require modifications to the core interferometer. The central Michelson interferometer is enclosed within vacuum tubes, so preserving the Michelson configuration avoids the need to alter the tube geometry. However, to manage two orthogonal polarizations, the substrate and coating properties of the mirrors must be modified, and considerable attention must be paid to the control of the multi-polarization system. Other designs, such as the sloshing speed meter²⁴ and Sagnac speed meter²⁵, utilize a single polarization but exploit spatial degrees of freedom to achieve backaction cancellation, as illustrated in Fig. 1b. These approaches require significant modifications to the central Michelson interferometer.

This work proposes a protocol to realize a speed meter through quantum teleportation. Quantum teleportation is widely regarded as a foundational technology for future quantum networks^{28,29} and distributed quantum computing systems³⁰. The reliable transfer of unknown quantum states between distant locations enables largescale quantum communication

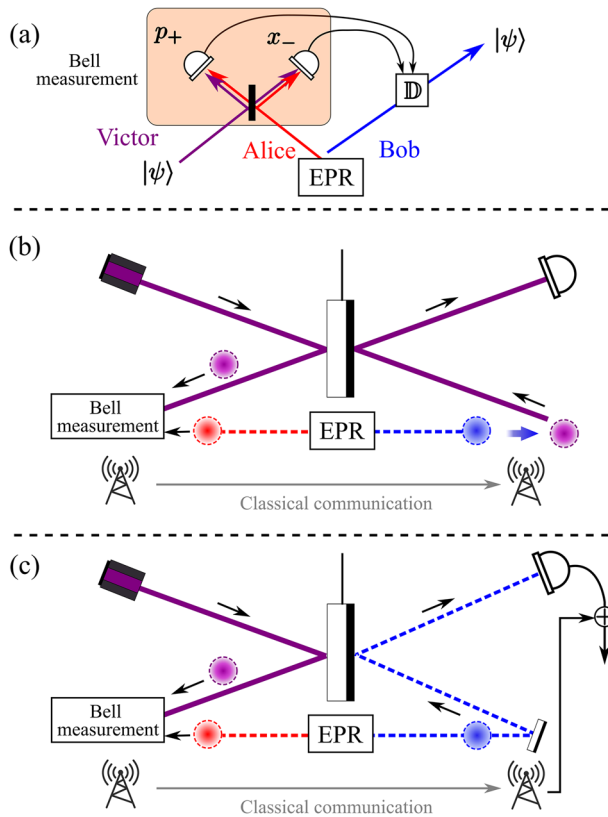


Fig. 2 | Continuous-variable teleportation protocol and speed-meter realization. **a** Diagram of continuous-variable teleportation. Alice (red) and Bob (blue) share an EPR-entangled state, and Victor (purple) prepares the target state $|\psi\rangle$. A Bell measurement on Victor's state and one of Alice's modes yields outcomes x_+ and p_+ , which Bob uses to apply a displacement operation to his mode. As a result, Victor's state is teleported to Bob. **b** Teleportation-based speed-meter interaction. Bob's mode is displaced online at the rear side of the mirror, and the teleported state interacts with the mirror a second time. **c** Bob's displacement is applied offline during post-processing. **b** and **c** are functionally equivalent and both realize a speed measurement.

protocols and significantly enhances data security^{31,32}. Quantum teleportation has also been proposed to extend the baseline of astrophysical telescopes³³.

A motivation of our scheme is similar to the conditional or teleportation broadband squeezing approaches proposed in Ref. 34,35, which aim to circumvent technical difficulties in GWDs by exploiting quantum entanglement. Those proposals eliminate the need for large-scale, low-loss, high-finesse filter cavities for broadband squeezing. Our proposal uses quantum entanglement to convert interferometric displacement sensors into speed meters without modifying the substrate or coating properties of the interferometer's core optics. In the case of the Michelson interferometer, for example, the front and rear light modes in Fig. 1b can be prepared in the frequency domain. By frequency-multiplexing the three parties-Victor, Alice, and Bob-the proposed protocol can be implemented without modifying the core optical configuration of the interferometer. This approach thus benefits both current^{36–38} and next-generation gravitational-wave detectors^{39,40}.

Results

Teleportation-based speed meter

Our teleportation scheme is based on the Braunstein-Kimble protocol⁴¹. In this protocol, the target state (hereafter referred to as *Victor*) is teleported from one half of an EPR pair (referred to as *Alice*) to the other half (referred to as *Bob*) via the transmission of classical information (see Fig. 2a). The transmitted information is obtained from a joint (Bell) measurement

performed on Alice and Victor. In an ideal EPR state, two modes are completely correlated, while the individual modes remain completely uncertain (i.e., they exhibit infinite EPR noise). In a conventional homodyne measurement, measuring one observable typically demolishes information about its conjugate; however, when the highly noisy EPR state is mixed with Victor's state prior to measurement, the measurement reveals no information about Victor's state. Due to the perfect quantum correlations in an ideal EPR state, only Bob can reconstruct Victor's state using the classical information from Alice.

In Fig. 2b, an Einstein-Podolsky-Rosen (EPR) entangled pair of photons is distributed to stations on the front and rear sides of a mirror, which we denote as Alice and Bob, respectively. The photon that interacts with the mirror on the front side is labeled Victor. Instead of recycling Victor to the mirror's rear side, Alice performs a joint (Bell) measurement on her photon and Victor's photon. The output of this Bell measurement is a combination of the two quadratures of the input fields necessary for teleportation.

Alice then transmits the measurement results to Bob via classical communication. Bob applies a displacement operation to his photon based on the received data, thereby teleporting Victor's state to himself. The teleported state is subsequently directed to the mirror from the rear side, where it interacts with the mirror once again. With perfect teleportation fidelity, this process is equivalent to the configuration illustrated in Fig. 1b. As shown later, the displacement process can be replaced by post-processing, as illustrated in Fig. 2c.

Sensitivity

In the lossless and infinite-squeezing limit, the input-output relation of the teleportation-based speed meter is given by

$$\hat{\mathbf{b}}_{\text{out}} = \mathbb{T}_v \hat{\mathbf{v}}_{\text{in}} + \frac{\mathbf{x}}{x_{\text{SQL}}} \mathbf{t} \quad (5)$$

where vectors $\hat{\mathbf{v}}_{\text{in}}$ and $\hat{\mathbf{b}}_{\text{out}}$ are the input and output states to the detector, and

$$\mathbb{T}_v = e^{4i\beta} \begin{pmatrix} 1 & 0 \\ -\mathcal{K}_{\text{sm}} & 1 \end{pmatrix} \quad (6)$$

$$\mathbf{t} = e^{2i\beta} \begin{pmatrix} 0 \\ \sqrt{2\mathcal{K}_{\text{sm}}} \end{pmatrix} \quad (7)$$

(see the Methods section). Here $\mathcal{K}_{\text{sm}}$ is defined as:

$$\mathcal{K}_{\text{sm}} = \frac{16\hbar\omega_a^2\gamma}{m(\gamma^2 + \Omega^2)^2}, \quad (8)$$

where ω_a is the optomechanical coupling constant and γ is the bandwidth of the arm cavity. The readout is performed by projecting the outgoing field onto the homodyne vector $\mathbf{H}_{\phi_{\text{LO}}}^T = \{\sin \phi_{\text{LO}}, \cos \phi_{\text{LO}}\}$:

$$\hat{\mathbf{b}}_{\phi_{\text{LO}}} = \mathbf{H}_{\phi_{\text{LO}}}^T \hat{\mathbf{b}}. \quad (9)$$

The spectral density for displacement measurement (displacement sensitivity) is then

$$S_x(\Omega) = x_{\text{SQL}}^2 \frac{\mathbf{H}_{\phi_{\text{LO}}}^T \mathbb{T}_v \mathbb{S}_v^{\text{in}} \mathbb{T}_v^\dagger \mathbf{H}_{\phi_{\text{LO}}}}{|\mathbf{H}_{\phi_{\text{LO}}}^T \mathbf{t}|^2} \quad (10)$$

The quantum spectral density of the input fields, $\mathbb{S}_\mu^{\text{in}}$, is defined by

$$2\pi\delta(\Omega - \Omega') \mathbb{S}_{v,j}^{\text{in}}(\Omega) \equiv \frac{1}{2} \left\langle \text{in} \left| \hat{v}_{i,\text{in}}(\Omega) \hat{v}_{j,\text{in}}^\dagger(\Omega') + \hat{v}_{j,\text{in}}^\dagger(\Omega') \hat{v}_{i,\text{in}}(\Omega) \right| \text{in} \right\rangle \quad (11)$$

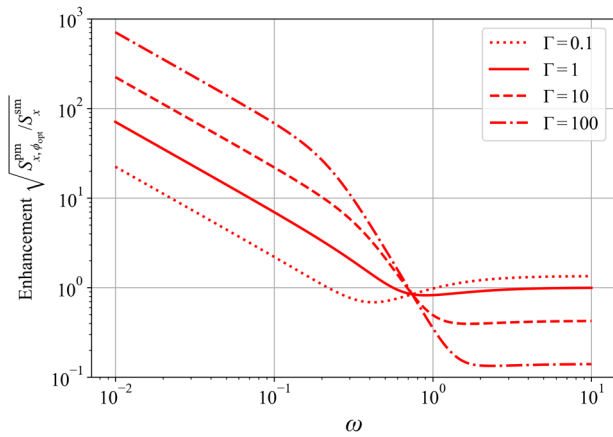


Fig. 3 | Enhancement in the sensitivity of the speed meter relative to the position meter for various values of Γ .

with $(i, j) = \{1, 2\}$ (see Refs. 23,26 for more details). For a readout angle of $\phi_{\text{LO}} = \pi/2$, the displacement sensitivity of the quantum noise-limited interferometer is given by

$$S_{x,\pi/2}^{\text{sm}} = \frac{x_{\text{SQL}}^2}{2} \left(\frac{1}{\mathcal{K}_{\text{sm}}} + \mathcal{K}_{\text{sm}} \right). \quad (12)$$

which agrees with the sensitivity shown in Ref. 26.

Enhancement

A key feature of the speed meter is its optomechanical coupling factor, $\mathcal{K}_{\text{sm}}$, which remains constant at low frequencies $\Omega \ll \gamma$. Consequently, the speed meter significantly reduces radiation pressure noise. Moreover, since $\mathcal{K}_{\text{sm}}$ remains constant at low frequencies, the radiation pressure noise can be canceled by exploiting correlations between the amplitude $\hat{b}_{1,\text{out}}$ and phase $\hat{b}_{2,\text{out}}$ quadratures without requiring frequency-dependent or variational readout (which would otherwise necessitate additional long filter cavities). The optimal homodyne (readout) angle at DC is given by

$$\phi_{\text{opt}} = \arccot[\mathcal{K}_{\text{sm}}(\Omega \rightarrow 0)]. \quad (13)$$

Thus, with a fixed-angle readout, the sensitivity can beat the SQL in the radiation-pressure-dominated region. In contrast, the coupling constant of the conventional position meter

$$\mathcal{K}_{\text{pm}} = \frac{4\gamma\hbar\omega_a^2}{m\Omega^2(\gamma^2 + \Omega^2)}, \quad (14)$$

exhibits a frequency dependence proportional to Ω^{-2} .

The enhancement factor, in the sensitivity of the speed meter compared to the position meter is given by

$$\frac{S_x^{\text{pm}}}{S_{x,\phi_{\text{opt}}}^{\text{sm}}} = \frac{2\omega^2 [1 + \omega^2(2 + \omega^2)\{2 + (1 + \Gamma)\omega^2(2 + \omega^2)\}]}{(1 + \omega^2)\{\Gamma + 4\omega^4(1 + \omega^2)^2\}}, \quad (15)$$

where

$$\Gamma = \frac{256\hbar^2\omega_a^4}{m^2\gamma^6}, \text{ and } \omega = \frac{\Omega}{\gamma} \quad (16)$$

(for the derivation of the position meter noise S_x^{pm} , see the supplementary Note I). Equation (15) converges to $(1 + \Gamma)/2$ in the limit $\omega \rightarrow \infty$, indicating that the high-frequency (shot-noise-limited) sensitivities of the speed meter and the position meter coincide when $\Gamma = 1$. Figure 3 shows the enhancement factor for several values of Γ . The enhancement at low

frequencies becomes more pronounced as Γ increases, which can be realized by increasing the laser power or reducing the mirror mass.

The enhancement beyond the SQL is quantified as

$$\rho^2 = \frac{x_{\text{SQL}}^2}{S_{x,\phi_{\text{opt}}}^{\text{sm}}} = 2\sqrt{\Gamma} \quad (17)$$

which is equivalent to the parametrization presented in Ref. 26. Thus, the speed meter beats the SQL when $\Gamma > \frac{1}{4}$.

Interferometric gravitational-wave detector

We apply the teleportation-based speed measurement scheme to a LIGO-type gravitational-wave detector and compare its displacement sensitivity to that of a conventional Michelson interferometer operating as a position meter, both using the same total circulating laser power. Panels (a) and (b) of Fig. 4 show the schematics of the online and offline approaches in a Michelson-type interferometer, respectively. The input fields, Bob and Victor, are injected through the Faraday isolator. Alice's and Victor's fields share the same frequency, ω_0 , while Bob's frequency is slightly detuned to $\omega_0 + \Delta\omega$. The pump fields at frequencies ω_0 and $\omega_0 + \Delta\omega$ co-resonate in the arm cavities. The beam splitters at the dark port combining the three beams are frequency-dependent; for example, a triangular optical cavity can be used to realize this. The coupling constant ω_a is defined as

$$\omega_a \equiv \sqrt{\frac{mJ}{2\hbar}}$$

with the normalized power $J = \frac{4P_c\omega_0}{mLc}$, where P_c is the total circulating power, ω_0 is the laser frequency, and c is the speed of light. In the online approach, a displacement operation is applied at the dark port, whereas in the offline approach, all operations are implemented via post-processing.

The left panel of Fig. 5 shows the displacement sensitivity curves in the lossless case. The system parameters are chosen to match those in Ref. 26, where the shot-noise-limited sensitivity of the position meter and the speed meter with the optimal homodyne angle are aligned to the same level by setting $\Gamma \sim 1$. At $\phi_{\text{LO}} = \pi/2$, the low-frequency noise nearly reaches the SQL, whereas using the optimal homodyne angle $\phi_{\text{LO}} = \phi_{\text{opt}}$ allows the sensitivity to surpass the SQL. Compared to the position meter, the amplitude sensitivity is improved by approximately one order of magnitude at 8 Hz.

The right panel of Fig. 5 shows the displacement sensitivity curves in the lossy case. Although losses degrade sensitivity, speed meters still yield a considerable enhancement at low frequencies compared to the Michelson position meters (shown in the solid-green curve). This is because the QND process of the speed meters cause inside the interferometer. Compared to that, readout-based schemes such as variational readout⁶ are more susceptible to losses (see section 6 in Ref. 23). Below 10 Hz, the sensitivity degrades due to uncorrelated vacuum noise introduced by losses. In our simulation, we assumed 2% input and 1% output losses for each of the three fields, $\mathcal{L} = 30\text{ppm}$ round-trip loss in the arm cavity, and path stability of the circulator $\Delta l \sim 10^{-4}\lambda_0$. These values are so ambitious but target values to be pursued in future detectors. The arm loss contributes as additional damping γ_2 as $\gamma_2 = \frac{\mathcal{L}c}{4L}$. The EPR entanglement is generated with 15 dB of squeezing-beyond which the sensitivity no longer improves because the interferometer losses dominate (note that 15 dB is the generated squeezing level, not the observed level). The low-frequency sensitivity scales as Ω^{-1} , reflecting the frequency dependence of the radiation pressure by the loss-induced vacuum. Despite the presence of losses, the low-frequency sensitivity remains enhanced, and with the optimal homodyne angle $\phi_{\text{LO}} = \phi_{\text{opt}}$ the sensitivity surpasses the SQL.

Finally, we compared the detector's performance with that of the polarization-circulation speed meter (PCSM), shown as the dashed blue curves in Fig. 5. The parameters for the PCSM follow those in Ref. 26, including imperfections such as 1% losses in the QWP and PBS, and the path-length fluctuation of the polarization circulator, $\Delta l \sim 10^{-4}\lambda_0$. In both

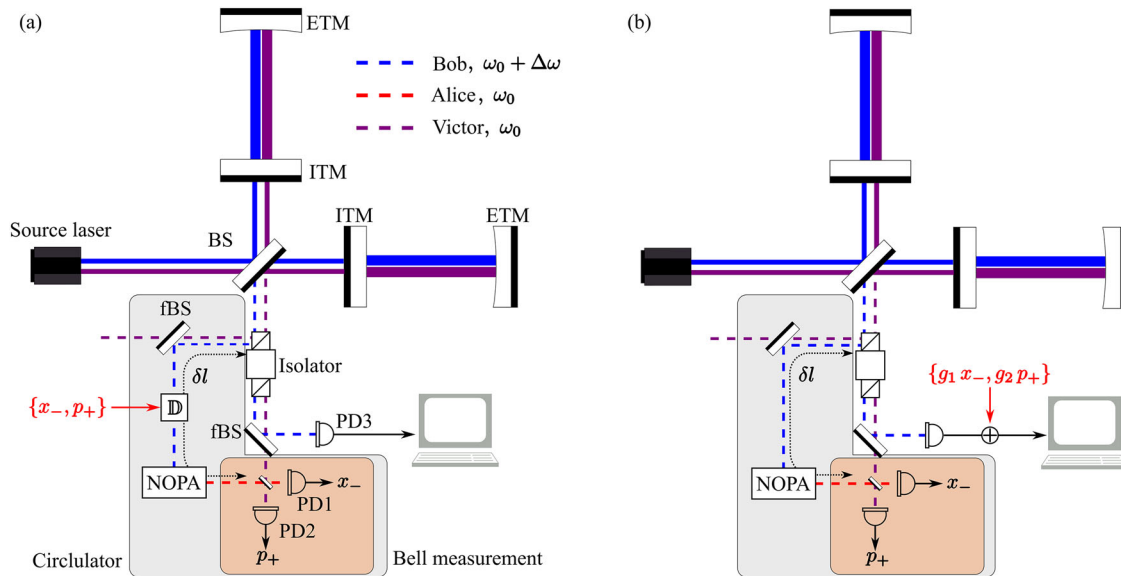


Fig. 4 | Schematics of the teleportation-based speed meter in a Michelson-type interferometer. a Online approach: Victor's field (purple) is combined with Alice's field (red) and sent to a Bell measurement. Based on the measurement outcome, Bob's field (blue) is displaced and then injected into the interferometer; its output is

subsequently detected by a homodyne detector. **b** Offline approach: the displacement operation is implemented offline through post-processing. ITM input test mass, ETM end test mass, BS beam splitter, fBS frequency beam splitter, NOPA non-degenerate optical parametric amplifier, PD photodetector.

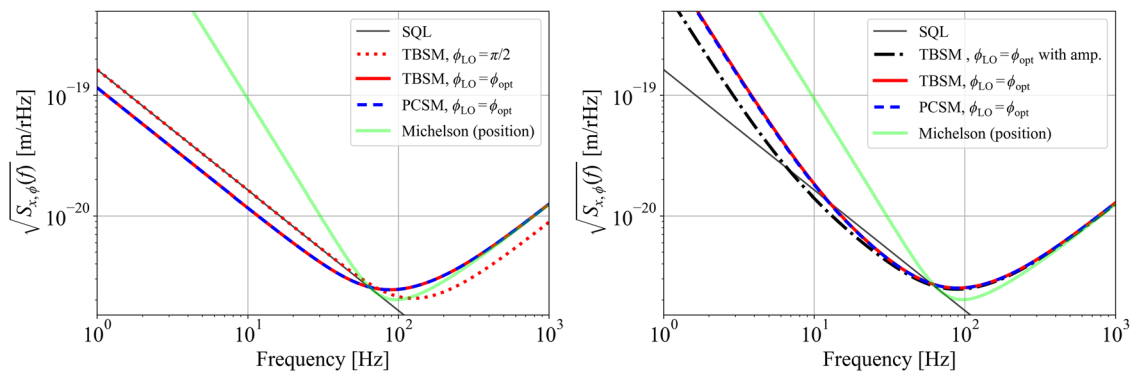


Fig. 5 | Displacement sensitivity of the speed meter applied to a gravitational-wave detector. Left panel: Sensitivity in the lossless limit, compared with the standard quantum limit (SQL). **Right panel:** Sensitivity in the presence of realistic losses. The solid red curves represent the teleportation-based speed meter (TBSM) with $\phi_{LO} = \phi_{opt}$, while the dotted red curves show results for $\phi_{LO} = \pi/2$. The dash-dotted black curve in the right panel shows the sensitivity of the TBSM when a readout amplifier is used. The solid green curve in both panels shows the sensitivity of a conventional Michelson interferometer used as a position meter, assuming no losses and no input squeezing. In the right panel, we assume input/output losses of $\epsilon_{in} = 2\%$ and $\epsilon_{out} = 1\%$, an arm cavity loss of 30 ppm, and 15 dB of generated two-

mode squeezing shared between Alice and Bob (Victor remains in vacuum) in the teleportation-based speed meter. The blue dashed curves represent the sensitivity of the polarization circulation speed meter (PCSM) in both the lossless (left) and lossy (right) scenarios. For the lossy case, we adopt the same loss parameters and additionally assume 1% loss in the polarization optics, such as the QWP and PBS 26. Both speed meter schemes include a circulator path fluctuation of $\Delta l \sim 10^{-4} \lambda_p$. Other parameters assumed throughout: arm length $L = 4$ km, mirror mass $m = 200$ kg, total circulating power $P_c = 3$ MW, laser wavelength $\lambda_p = 2 \mu\text{m}$, and cavity bandwidth $\gamma = 2\pi \times 115 \text{ Hz}^{26}$.

the lossless and lossy cases shown, the sensitivity curves of the two speed meter configurations closely overlap. This result shows that the teleportation-based speed meter is a promising candidate for future detectors, not only due to its ease of implementation but also its sensitivity performance.

Discussion

In the online approach, the teleportation works as a frequency converter, which is analogous to the polarization circulator in Ref. 26. The benefit of using quantum teleportation is that this operation can instead be achieved offline, which is not possible with a classical frequency converter. The measurement process mediated by quantum teleportation can be applied to a wide range of precision sensors,

including Michelson interferometers, Mach-Zehnder interferometers, or atom interferometers.

When applied to interferometric gravitational-wave detectors, our scheme requires no modifications to the core interferometer optics including the coating properties, only by altering the dark-port components and introducing a second pump from the bright port (which is similar to the paired carrier interferometry in Ref. 42,43). The teleportation-based speed meter needs only two pumping lasers at slightly different frequencies (detuned on the order of MHz). Such two-color lasers can be generated, for example, by using an electro-optic modulator.

The simulation presented omits certain experimental details of the teleportation procedure. In the online approach, the physical displacement operation —implemented using a low-transmissivity mirror and a

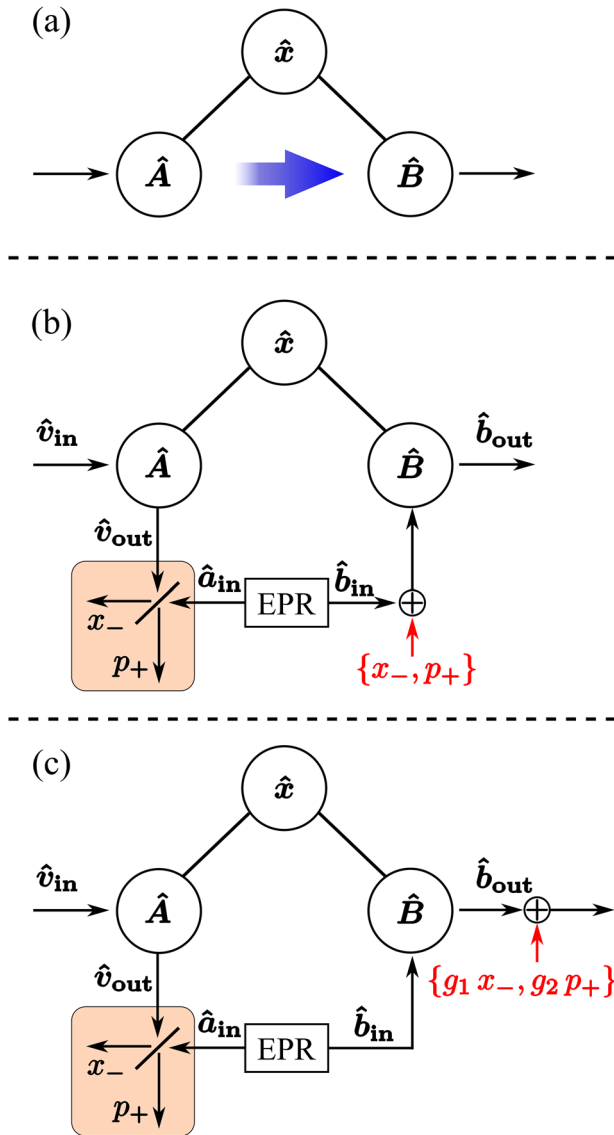


Fig. 6 | Mode and Input-Output Diagrams of the teleportation-based speed meter. **a** A mode diagram of the speed meter. $\hat{x}$ denotes the position operator of the test mass, while $\hat{A}$ and $\hat{B}$ are the cavity modes that couple to $\hat{x}$. By coupling modes $\hat{A}$ and $\hat{B}$ nonreciprocally (blue arrow) and flipping the signs of the interaction Hamiltonians between $\hat{x}$ and $\hat{A}/\hat{B}$, a speed measurement can be realized. Detailed input-output relations of the teleportation-based speed meter in the online (b) and offline (c) approaches. The orange shaded areas represent Bell measurement.

modulated coherent laser⁴⁴—may introduce additional losses. Moreover, although not critical under current conditions where the arm cavity bandwidth is approximately 100 Hz, the latency associated with the teleportation feedforward process could become problematic in relatively broadband detectors targeting higher frequencies. These practical considerations make the offline approach more favorable for implementation.

A major challenge in entanglement-based schemes is to mitigate the noise contribution from input and output losses. In our approach, three detection ports contribute output noise (a threefold effect), while input losses affect both Victor and Bob (a twofold effect). The readout noise can be mitigated by employing a readout amplifier^{45–48}. The dashed curve in the right panel of Fig. 5 shows that readout amplification improves sensitivity in the radiation-pressure-dominated band, compared to that without the amplifier. Readout

amplification is crucial not only for the teleportation-based speed meter but also for other multi-color interferometric techniques.

In this article, we propose an alternative speed meter scheme for interferometric displacement sensors based on quantum teleportation. We introduce two equivalent implementations—an online approach and an offline approach—both of which achieve quantum noise reduction below the standard quantum limit (SQL), even in the presence of losses. When applied to interferometric gravitational-wave detectors, the scheme requires no modification to the core interferometer design or to mirror properties such as coatings or substrate materials. This provides a practical route to realizing speed meter operation, evading back-action noise, and improving low-frequency sensitivity.

Methods

We first outline the preparation of the initial states for teleportation, which is the key element of continuous variable teleportation illustrated in Fig. 2a. Next, we analyze the system dynamics of the online and offline approaches using a general Hamiltonian model. Finally, we examine the impact of imperfections.

State preparation and system dynamics

Throughout this work, we adopt a quadrature representation based on the two-photon formalism^{49,50}. Given that $\hat{a}_\omega$ denotes the single-photon annihilation operator of the mode with frequency ω , the two-photon quadrature amplitudes are defined as:

$$\hat{a}_{1,\Omega} = \frac{\hat{a}_{\omega_0+\Omega} + \hat{a}_{\omega_0-\Omega}^\dagger}{\sqrt{2}} \quad (18)$$

$$\hat{a}_{2,\Omega} = \frac{\hat{a}_{\omega_0+\Omega} - \hat{a}_{\omega_0-\Omega}^\dagger}{i\sqrt{2}} \quad (19)$$

where ω_0 is the carrier frequency and Ω is the sideband frequency. We denote the fields associated with Victor, Alice, and Bob as

$$\hat{a} = \begin{pmatrix} \hat{a}_{1,\Omega} \\ \hat{a}_{2,\Omega} \end{pmatrix}, \quad \hat{b} = \begin{pmatrix} \hat{b}_{1,\Omega} \\ \hat{b}_{2,\Omega} \end{pmatrix}, \quad \hat{v} = \begin{pmatrix} \hat{v}_{1,\Omega} \\ \hat{v}_{2,\Omega} \end{pmatrix}. \quad (20)$$

For brevity, we omit the explicit Ω dependence hereafter and introduce subscripts “in” and “out” to denote input and output fields, for example, $\hat{a}_{\text{out}} = \{\hat{a}_{1,\text{out}}, \hat{a}_{2,\text{out}}\}^T$.

The EPR entanglement, which is a two-mode squeezed vacuum state, is characterized in terms of the quantum spectral density of the four EPR operators $(\hat{a}_{j,\text{in}} \pm \hat{b}_{j,\text{in}})/\sqrt{2}$ for $j = 1, 2$ as⁵¹:

$$S_{(\hat{a}_{j,\text{in}} \pm \hat{b}_{j,\text{in}})/\sqrt{2}} = e^{\mp(-1)^j 2r} \quad (21)$$

where r is the squeezing factor. When $r \rightarrow \infty$, the noise spectra of $S_{(\hat{a}_1 - \hat{b}_1)/\sqrt{2}}$ and $S_{(\hat{a}_2 + \hat{b}_2)/\sqrt{2}}$ approach zero, corresponding to the original EPR entanglement⁵².

The input field for Victor, $\hat{v}_{\text{in}}$ (a pure vacuum state, depicted in purple), is injected to mode $\hat{A}$, interacts with the pump, and then exits as $\hat{v}_{\text{out}}$. This field enters the Bell measurement and results in measurement outcomes as follows:

$$\alpha = \frac{1}{\sqrt{2}} \begin{pmatrix} v_{1,\text{out}} - a_{1,\text{in}} \\ v_{2,\text{out}} + a_{2,\text{in}} \end{pmatrix} \equiv \begin{pmatrix} x_- \\ p_+ \end{pmatrix}. \quad (22)$$

We now describe the system dynamics using analytic Hamiltonian formalism. Figure 6a illustrates the mode diagram of the non-reciprocal speed measurement. Two cavity modes, $\hat{A}$ and $\hat{B}$, are

coupled to the position of the oscillator's mechanical mode $\hat{x}$. The interaction between $\hat{A}$ and $\hat{B}$ is non-reciprocal: the outgoing field from $\hat{A}$ enters $\hat{B}$, while the reverse process does not take place. As a result, the input field entering $\hat{A}$ exits from $\hat{B}$ after interacting with the mirror twice. By inverting the sign of the two terms in the Hamiltonian related to the optomechanical interaction, the measurement's back-action can be effectively canceled out. The main objective of this study is to realize the configuration illustrated in this diagram through quantum teleportation.

The input and output relation of the fields and modes is depicted in Fig. 6a. The Hamiltonian of the system is given by^{33,34}:

$$\hat{H} = \hbar\omega_a(\hat{A} + \hat{A}^\dagger)(\hat{x} - x) - \hbar\omega_a(\hat{B} + \hat{B}^\dagger)(\hat{x} - x) + \frac{1}{2}\left(\frac{\hat{p}^2}{m} + m\omega_m^2\hat{x}^2\right) \quad (23)$$

where x is the displacement by external classical force, m is the mirror mass, and ω_m and $\hat{p}$ denote the natural frequency and momentum operator of the mechanical mode, respectively. ω_a is the optomechanical coupling constant between the mechanical and cavity modes. In deriving Eq. (23), we have applied the rotating-wave approximation to ignore the self-evolution of the light modes.

The Heisenberg-Langevin equations of motion in the time domain are thus:

$$\begin{aligned} \frac{d\hat{A}}{dt} &= -\gamma\hat{A} + \sqrt{2\gamma}\hat{v}_{\text{in}} - i\omega_a(\hat{x} - x) \\ \frac{d\hat{B}}{dt} &= -\gamma\hat{B} + \sqrt{2\gamma}\hat{b}_{\text{in}} + i\omega_a(\hat{x} - x) \\ \frac{d\hat{x}}{dt} &= \frac{\hat{p}}{m} \\ \frac{d\hat{p}}{dt} &= -\hbar\omega_a(\hat{A} + \hat{A}^\dagger - \hat{B} - \hat{B}^\dagger) - m\omega_m^2\hat{x} \end{aligned} \quad (24)$$

In the Fourier domain and expressed in terms of quadrature operators, these equations become:

$$\begin{aligned} -i\Omega\hat{A}_1 &= -\gamma\hat{A}_1 + \sqrt{2\gamma}\hat{v}_{1,\text{in}} \\ -i\Omega\hat{A}_2 &= -\gamma\hat{A}_2 + \sqrt{2\gamma}\hat{v}_{2,\text{in}} - \sqrt{2}\omega_a(\hat{x} - x) \\ -i\Omega\hat{B}_1 &= -\gamma\hat{B}_1 + \sqrt{2\gamma}\hat{b}_{1,\text{in}} \\ -i\Omega\hat{B}_2 &= -\gamma\hat{B}_2 + \sqrt{2\gamma}\hat{b}_{2,\text{in}} - \sqrt{2}\omega_a(\hat{x} - x) \\ -i\Omega\hat{x} &= \frac{\hat{p}}{m} \\ -i\Omega\hat{p} &= -\sqrt{2}\hbar\omega_a(\hat{A}_1 - \hat{B}_1) - m\omega_m^2\hat{x} \end{aligned} \quad (25)$$

where γ is the cavity's bandwidth. Here, for $X = (\hat{A}, \hat{B})$, we define the quadratures as

$$X_1 = \frac{X + X^\dagger}{\sqrt{2}}, \quad X_2 = \frac{X - X^\dagger}{i\sqrt{2}} \quad (26)$$

and denote $\mathbf{X} = \{X_1, X_2\}^T$. The output fields of the two cavity modes are given by:

$$\begin{aligned} \hat{v}_{\text{out}} &= -\hat{v}_{\text{in}} + \sqrt{2\gamma}\hat{A}, \\ \hat{b}_{\text{out}} &= -\hat{b}_{\text{in}} + \sqrt{2\gamma}\hat{B} \end{aligned} \quad (27)$$

which are the standard input-output relations.

Online approach

In the "online" approach, the displacement operation is applied to the light field in real time. Here, we first analyze the direct implementation as it provides clearer physical intuition and directly corresponds to the schematic in Fig. 6b. We will later show that this operation can be replaced with post-processing (which we call the "offline" approach).

Bob's field, denoted as $\hat{v}_{\text{out}}$, is displaced based on the Bell measurement's outcomes $\{x_-, p_+\}^T$ and then injected into the mode $\hat{B}$. The field subsequently couples with $\hat{x}$, which cancels the back-action, and then emerges as the output as $\hat{b}_{\text{out}}$.

The displacement operation, based on the Bell measurement Eq. (22), is expressed as

$$\mathbb{D}[\hat{b}_{\text{in}}; x_-, p_+] = \hat{b}_{\text{in}} + \sqrt{2}\alpha = \begin{pmatrix} \hat{b}_{1,\text{in}} + x_- \\ \hat{b}_{2,\text{in}} + p_+ \end{pmatrix}. \quad (28)$$

Using Eq. (21), Eq. (28) can be rewritten as

$$\hat{b}_{\text{in}} = \sqrt{2}e^{-r}\hat{z} + \hat{v}_{\text{out}} \quad (29)$$

where the operators $\hat{z}_{1,2}$ have unit variance (i.e., $S_{z_1} = S_{z_2} = 1$) and we define $\hat{z} = \{\hat{z}_1, \hat{z}_2\}^T$.

The output field from mode $\hat{B}$ is then given by

$$\hat{b}_{\text{out}} = \mathbb{T}_v\hat{v}_{\text{in}} + \sqrt{2}e^{-r}\mathbb{T}_z\hat{z} + \frac{x}{x_{\text{SQL}}}\mathbf{t}, \quad (30)$$

where

$$\mathbb{T}_z = e^{2i\beta} \begin{pmatrix} 1 & 0 \\ -\mathcal{K}_z & 1 \end{pmatrix} \quad (31)$$

and $\mathbb{T}_v$ and $\mathbf{t}$ are given by Eqs. (6) and (7). Here, $\beta = \arctan \frac{\Omega}{\gamma}$ is a phase rotation common to both quadratures, and $\mathcal{K}_{\text{sm}}$ (given in Eq. (8)) and $\mathcal{K}_z$ are the optomechanical coupling factors for the speed meter and the auxiliary field $\hat{z}$, respectively:

$$\mathcal{K}_z = \frac{\gamma + i\Omega}{2i\Omega}\mathcal{K}_{\text{sm}} \quad (32)$$

The spectral density is then

$$S_x(\Omega) = x_{\text{SQL}}^2 \frac{\sum_\mu \mathbf{H}_{\phi_{\text{LO}}}^T \mathbb{D}_\mu S_\mu^{\text{in}} \mathbb{D}_\mu^\dagger \mathbf{H}_{\phi_{\text{LO}}}}{|\mathbf{H}_{\phi_{\text{LO}}}^T \mathbf{t}|^2} \quad (33)$$

where $\mu \in \{v, z\}$. For a readout angle of $\phi_{\text{LO}} = \pi/2$ and in the limit $r \rightarrow \infty$, and with the definition of spectral density given by Eq. (11), the displacement sensitivity of the quantum noise-limited interferometer is given by Eq. (12).

Offline approach

The online displacement operation can be replaced by post-processing, which is experimentally easier as it can be implemented "offline". The

schematic diagram is shown in Fig. 6c. In this case, the input-output relation can be obtained by setting $x_- = p_+ = 0$ in Eq. (28) as:

$$\hat{\mathbf{b}}_{\text{out}} = \mathbb{T}_{\text{b}} \hat{\mathbf{b}}_{\text{in}} + \mathbb{T}'_{\text{v}} \hat{\mathbf{v}}_{\text{in}} + \frac{x}{x_{\text{SQL}}} \mathbf{t}' \quad (34)$$

with

$$\mathbb{T}_{\text{b}} = e^{2i\beta} \begin{pmatrix} 1 & 0 \\ -\mathcal{K}_{\text{pm}} & 1 \end{pmatrix} \quad (35)$$

$$\mathbb{T}'_{\text{v}} = e^{2i\beta} \begin{pmatrix} 0 & 0 \\ \mathcal{K}_{\text{pm}} & 0 \end{pmatrix} \quad (36)$$

$$\mathbf{t}' = e^{i\beta} \begin{pmatrix} 0 \\ \sqrt{2\mathcal{K}_{\text{pm}}} \end{pmatrix} \quad (37)$$

Here, $\mathcal{K}_{\text{pm}}$ is the optomechanical coupling constant for the position meter defined in Eq. (14) which satisfies the relation:

$$|\mathcal{K}_z|^2 = \mathcal{K}_{\text{sm}} \mathcal{K}_{\text{pm}} \quad (38)$$

For simplicity, in the limit $r \rightarrow \infty$, the measured output $b_{2,\text{out}}$ (with $\phi_{\text{LO}} = 0$) is conditioned by combining it with α using Wiener filters g_1 and g_2 :

$$b_{2,\text{out}}^{\text{con}} = b_{2,\text{out}} + g_1 x_- + g_2 p_+ \quad (39)$$

The optimal filters are found to be

$$g_1 = e^{2i\beta} \text{ and } g_2 = -\mathcal{K}_z^* e^{2i\beta} \quad (40)$$

under which the offline displacement sensitivity agrees with the online value in Eq. (12).

Compared to the online approach or ordinary speed meter schemes, the offline approach has the distinct feature that the back-action force acting on the mirror is not actually canceled. In the toy model of Fig. 2c, the backaction force exerted by Victor's field on the front side of the mirror has no correlation with the force by Bob's field, so ordinary homodyne detection fails to cancel it. Nevertheless, the back-action is canceled in the eventual sensitivity curve, conditionally erased in post-processing. The key is the Bell measurement, which creates tripartite entanglement among the EPR pair and Victor⁴¹, thereby effectively enabling the cancellation of the backaction force in the offline data.

Imperfection analysis

We now incorporate four imperfections into the sensitivity analysis: arm cavity losses (γ_2), input losses (ϵ_{in}), output losses (ϵ_{out}), and path length deviations (Δl) in the circulator (see Fig. 4 for the location of the circulator). The following analysis is intended only as a guideline for evaluating their impact.

Losses inside the arm cavity introduce extra vacuum noise into the equations of motion:

$$-i\Omega \hat{A}_1 = -(\gamma + \gamma_2) \hat{A}_1 + \sqrt{2\gamma} \hat{v}_{1,\text{in}} + \sqrt{2\gamma_2} \hat{a}'_1$$

$$-i\Omega \hat{A}_2 = -(\gamma + \gamma_2) \hat{A}_2 + \sqrt{2\gamma} \hat{v}_{2,\text{in}} - \sqrt{2\omega_a} (\hat{x} - x) + \sqrt{2\gamma_2} \hat{a}'_2$$

$$-i\Omega \hat{B}_1 = -(\gamma + \gamma_2) \hat{B}_1 + \sqrt{2\gamma} \hat{b}_{1,\text{in}} + \sqrt{2\gamma_2} \hat{b}'_1$$

$$-i\Omega \hat{B}_2 = -(\gamma + \gamma_2) \hat{B}_2 + \sqrt{2\gamma} \hat{b}_{2,\text{in}} - \sqrt{2\omega_a} (\hat{x} - x) + \sqrt{2\gamma_2} \hat{b}'_1 \quad (41)$$

Here, the loss-induced vacuum fields ($\hat{\mathbf{a}}' = \{\hat{a}'_1, \hat{a}'_2\}^T$ and $\hat{\mathbf{b}}' = \{\hat{b}'_1, \hat{b}'_2\}^T$) have unit covariance, and γ_2 is the mode extraction rate due to the power loss.

Additional losses occur in the detection and input optics. The output fields become:

$$\hat{\alpha}^{\text{L}} = \sqrt{1 - \epsilon_{\text{out}}} \mathbb{R}(\delta\theta) \hat{\alpha} + \sqrt{\epsilon_{\text{out}}} \hat{\alpha}', \quad (42)$$

$$\hat{\mathbf{b}}_{\text{out}}^{\text{L}} = \sqrt{1 - \epsilon_{\text{out}}} \hat{\mathbf{b}}_{\text{out}} + \sqrt{\epsilon_{\text{out}}} \hat{\mathbf{b}}'_{\text{out}} \quad (43)$$

and the input Bob's field is modified as:

$$\hat{\mathbf{b}}_{\text{in}}^{\text{L}} = \sqrt{1 - \epsilon_{\text{in}}} \hat{\mathbf{b}}_{\text{in}} + \sqrt{\epsilon_{\text{in}}} \hat{\mathbf{b}}'_{\text{in}}. \quad (44)$$

Here $\hat{\alpha}' = \{\hat{\alpha}'_1, \hat{\alpha}'_2\}^T$, $\hat{\mathbf{b}}'_{\text{in}} = \{\hat{b}'_{1,\text{in}}, \hat{b}'_{2,\text{in}}\}^T$ and $\hat{\mathbf{b}}'_{\text{out}} = \{\hat{b}'_{1,\text{out}}, \hat{b}'_{2,\text{out}}\}^T$ are the loss-induced vacuum fields with unit covariance. $\mathbb{R}(\delta\theta)$ is a rotation matrix with $\delta\theta = 2\pi\delta l/\lambda_0$, where $\lambda_0 = c/(\omega_0/2\pi)$ denotes the wavelength of the carrier light, and δl represents the instantaneous path length fluctuation in the circulator. The statistical properties of δl are given by $\delta l = 0$ and $\langle \delta l^2 \rangle = \Delta l^2$, where Δl is the root-mean-square (RMS) value of δl . This matrix captures the unintended phase rotation imparted to the optical fields of Alice and Victor, effectively modeling the path-length fluctuations in the polarization circulator as described in Ref. 26.

Using the transfer matrices for each optical field, the overall displacement sensitivity can be computed. By substituting Eq. (29) into Eq. (44) for the online and Eq. (34) into Eq. (43) for the offline approaches, we derive results that are identical for both approaches, as presented in Supplementary Note II.

Data availability

No datasets were generated or analyzed during the current study.

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Author contributions

Y.N. and J.G. conceived the idea and Y.C. and K.S. shaped it. Y.N. wrote the manuscript and J.G., Y.C. and K.S. reviewed and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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